

Cohort Section: CSE 3-A

Rule 0: Year 3 (Lunch Slot 5)

🚀 Run Master 17-Rule Solver

DAY / SLOT	SLOT 1	SLOT 2	SLOT 3	SLOT 4	SLOT 5 (LUNCH)	SLOT 6	SLOT 7
Monday	COMPILER DESIGN Prof. Dr. P. Physics I-506	COMPILER DESIGN Prof. Dr. K. Srinivasa Rao I-506	COMPILER DESIGN Prof. Dr. K. Srinivasa Rao I-506	COMPUTER NETWORKS Prof. Dr. K. Srinivasa Rao I-506	LUNCH BREAK	COMPUTER NETWORKS Prof. Dr. K. Srinivasa Rao C-301	COMPUTER NETWORKS Prof. Dr. V. Murali Krishna C-301
Tuesday	+ Assign	+ Assign	+ Assign	+ Assign	LUNCH BREAK	+ Assign	+ Assign
Wednesday	+ Assign	+ Assign	+ Assign	+ Assign	LUNCH BREAK	+ Assign	+ Assign
Thursday	+ Assign	+ Assign	+ Assign	+ Assign	LUNCH BREAK	+ Assign	+ Assign
Friday	+ Assign	+ Assign	+ Assign	+ Assign	LUNCH BREAK	+ Assign	+ Assign
Saturday	+ Assign	+ Assign	+ Assign	+ Assign	LUNCH BREAK	HALF DAY	HALF DAY

Real-Time Telemetry Simulator & Synthetic Fault Injector

Stream AI4I 2020 dataset telemetry in real time or trigger synthetic failure degradation scenarios.

Stream Interval: 1.0s (Normal)

Start Stream

Synthetic Fault Injection Suite (Trigger Degradation Scenario):

- Inject Tool Wear Failure (TWF)
- Inject Heat Dissipation Failure (HDF)
- Inject Power Failure (PWF)
- Inject Overstrain Failure (OSF)

Fleet Health Index

66.7%

Overall health score across registered assets

Active Failure Alarms

1

1 Critical | 0 Warning

Downtime Cost Savings

\$1,65,000

Prevented breakdown loss vs PM cost

Active Assets

3

Multi-tenant milling machines monitored

Equipment Assets Registry

Real-time telemetry & failure risk status

CNC Milling Machine #1

Product Type: M

3-Axis Vertical CNC Milling Center

Health Score 0.1%

Failure Risk

99.9%

Inspect Details

Precision Lathe #2

Product Type: L

High Precision Turning Lathe

Health Score 100.0%

Failure Risk

0.0%

Inspect Details

Heavy Industrial Mill #3

Product Type: H

Heavy Duty Production Milling Asset

Health Score 100.0%

Failure Risk

0.0%

Inspect Details

[▶ Start Stream](#)

Inject Overstrain Failure (OSF)

 Refresh Agent



PredictSense

Predictive Maintenance Profiler & AI Agent



Fleet Overview



Machine Inspector



Alert Triage



Maintenance ROI



Knowledge SOPs



Stream Paused

Start Live Telemetry

Real-Time Telemetry Simulator & Synthetic Fault Injector

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Stream Interval: 1.0s (Normal)



Start Stream

Synthetic Fault Injection Suite (Trigger Degradation Scenario):

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Alert Triage & Priority Feed

Cost-weighted prioritized maintenance alert queue

ALL

ACTIVE

ACKNOWLEDGED



CNC Milling Machine #1

CRITICAL

Priority: 100.0

High Risk Alert: OSF predicted with 100% probability on CNC Milling Machine #1.

Acknowledge

Triggered: 4:44:59 am Type: OSF

Synthetic Fault Injection Suite (Trigger Degradation Scenario):

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RAG Knowledge Base & SOP Manuals

 Search SOP procedures...

Tool Wear Failure (TWF) SOP

Cutting Tool Wear

Heat Dissipation Failure (HDF) SOP

Thermal Chiller System

Power Failure (PWF) SOP

Electrical & Transmission Drive

Overstrain Failure (OSF) SOP

Mechanical Strain

Random Failure (RNF) SOP

Stochastic & Electrical Surges

Tool Wear Failure (TWF) SOP

CUTTING TOOL WEAR

Trigger Telemetry Threshold:

Tool Wear > 200 min | Torque +15%

Action Protocol Steps:

1. Pause CNC cycle at tool retract position.
2. Inspect cutting edge for micro-chipping or crater wear.
3. Unclamp tool holder & insert fresh carbide insert.
4. Zero tool offset Z-axis precision ± 0.005 mm.

 Cost: \$150-\$350 | Downtime: 15-20 mins